

• Let A be an invertible matrix. By defn, there exists a unique $n \times n$ matrix A^{-1} s.t. $A^{-1}A = AA^{-1} = I$.

$$\begin{aligned} S(T(x)) &= S(Ax) = A^{-1}(Ax) = (A^{-1}A)x \quad (\text{Theorem 2(a)}) \\ &= Ix \quad (\text{Defn. of inverse}) \\ &= I \quad (\text{Defn. of identity matrix}) \end{aligned}$$

$$\begin{aligned} T(S(x)) &= T(A^{-1}x) = A(A^{-1}x) = (AA^{-1})x \quad (\text{Theorem 2(a)}) \\ &= Ix \quad (\text{Defn. of inverse}) \\ &= I \quad (\text{Defn. of identity matrix}) \end{aligned}$$

Verification of $A^{-1}x = S(x)$ is a linear transformation

Condition 1: $S(x+y) = A^{-1}(x+y) = A^{-1}x + A^{-1}y$ (Theorem 5(c) from Chapter 1)
 $= S(x) + S(y)$

Condition 2: Let c be an arbitrary scalar.
 $S(cx) = A^{-1}(cx) = c(A^{-1}x)$ (Theorem 5(b) from Chap-1)
 $= cS(x)$

Thus T is invertible. The proof that S is unique is outlined in Exercise 39. $\square$

Example 2: What can you say about a one-to-one linear transformation T from $\mathbb{R}^n$ into $\mathbb{R}^n$?

Solution: By Theorem 11 from Chapter 1, the equation $T(x) = 0$ has only the trivial solution. By Invertible Matrix Theorem and $\Leftrightarrow$, A is the standard matrix of T is invertible. By Theorem 9, T is invertible.

Practice Problems

1. Determine if $A = \begin{bmatrix} 2 & 3 & 4 \\ 2 & 3 & 4 \\ 2 & 3 & 4 \end{bmatrix}$ is invertible.

Solution:

$$\begin{bmatrix} 2 & 3 & 4 \\ 2 & 3 & 4 \\ 2 & 3 & 4 \end{bmatrix}$$

$$\begin{array}{l} \text{Row 2} = \text{Row 2} - \text{Row 1} \\ \text{Row 3} = \text{Row 3} - \text{Row 1} \end{array}$$

$$\begin{bmatrix} 2 & 3 & 4 \\ 0 & 0 & 0 \\ 2 & 3 & 4 \end{bmatrix}$$

$$\begin{array}{l} \text{Row 1} = \text{Row 1} \times 1/2 \\ \text{Row 3} = \text{Row 3} - \text{Row 1} \end{array}$$

$$\begin{bmatrix} 1 & 3/2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{array}{l} \text{Row 3} = \text{Row 3} - \text{Row 1} \\ \text{Row 2} = \text{Row 2} - \text{Row 1} \end{array}$$

$$\begin{bmatrix} 2 & 3 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$